



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

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SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

Fax: +66(0)-2721-3515

1. CHEMICAL IDENTIFICATION

Product Identification

Trade name: Domark

Product name: Tetraconazole 4% EW

Product use: Fungicide

Relevant identified uses of the substance/mixture and uses advised against:

Commonly used in agriculture and horticulture. Other use not admitted.

2. HAZARD IDENTIFICATION

Adverse Human Health Effects:

Inhalation,

Primary route of exposure.

Eye contact,

Direct contact with material can cause the following: moderate eye irritation. Avoid contact with eyes or clothing.

Skin contact,

Primary route of exposure.

Ingestion,

Harmful if swallowed.

3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name: (RS)-2-(2,4-dichlorophenyl)-3-(1H-1,2,4-triazol-1-yl)-propyl-1,1,2,2-tetrafluoroethyl ether

Common name: Tetraconazole (proposed to ISO)

Chemical class: Azole

Empirical formula: C₁₃H₁₁Cl₂F₄N₃O



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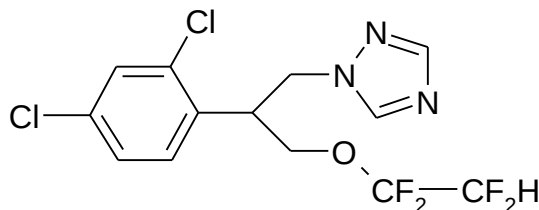
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Molecular mass: 372.14 g/mol

Structural formula:



Xn,N; R20/22-51/53

4.1/C2 Aquatic Chronic 2 H411

3.1/4/Oral Acute Tox. 4 H302

3.1/4/Inhal Acute Tox. 4 H332

1% - 3% Sodium dioctyl sulfosuccinate

CAS: 577-11-7

Xi; R36/38

3.2/2 Skin Irrit. 2 H315

3.3/2 Eye Irrit. 2 H319

Ingredients and composition:

Component	CAS No.	% W/V
Active ingredient (Tetraconazole)		
- As pure a.i.	112281-77-3	4
- As technical a.i.		4.3
Emulsifiers	-	21
Deionized water	-	To make 1 L

4. FIRST AID MEASURES

Description of first aid measures

In case of skin contact:

Wash with plenty of water and soap.

In case of eyes contact:

In case of contact with eyes, rinse immediately with plenty of water for at least 10 minutes and seek medical advice.

In case of Ingestion:

Do not under any circumstances induce vomiting. OBTAIN A MEDICAL EXAMINATION IMMEDIATELY and present the safety-data sheet.

A suspension of activated charcoal in water, or liquid paraffin may be administered.

In case of Inhalation:

Ventilate the premises. The patient is to be removed immediately from the contaminated premises to fresh air, keep warm and made to rest in a well ventilated area. If the patient feel unwell, OBTAIN MEDICAL ATTENTION.

Most important symptoms and effects, both acute and delayed

None

Indication of any immediate medical attention and special treatment needed Treatment:

None



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5. FIRE FIGHTING MEASURES

Flash point & Method:	> 100.5 °C (EEC A.9)
Upper/lower flammability or explosive limits:	Not available
Auto-ignition temperature:	450 °C
Decomposition temperature:	Not available
Explosive properties:	Not explosive (on the basis of components)
Oxidizing properties:	Not oxidizing (on the basis of the components)

FIRE FIGHTING HAZARDS& PROCEDURES:

Extinguishing media

Suitable extinguishing media:

Water

Carbon dioxide (CO₂)

Extinguishing media which must not be used for safety reasons:

None in particular

Special hazards arising from the substance or mixture

Do not inhale explosion and combustion gases which, at high temperatures, may contain toxic substances such as CO_x, NO_x, HCl, HF and HCN. Burning produces heavy smoke.

Advice for fire-fighters

Use suitable breathing apparatus. Self-contained breathing apparatus.

Collect contaminated fire extinguishing water separately. This must not be discharged into drains.

Move undamaged containers from immediate hazard area if it can be done safely.

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Personal precautions, protective equipment and emergency procedures

Wear personal protection equipment.

Remove persons to safety.

See protective measures under point 7 and 8.

Environmental precautions

Do not allow to enter into soil/subsoil. Do not allow to enter into surface water or drains. Retain contaminated washing water and dispose it.

In case of gas escape or of entry into waterways, soil or drains, inform the responsible authorities.

Suitable material for taking up: absorbing material, organic, sand



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Methods and material for containment and cleaning up

Large Spills:

Stop the flow of material, if this is without risk. Dike the spilled material, where this is possible.

Cover with plastic sheet to prevent spreading. Absorb in vermiculite, dry sand or earth and place into containers. Follow product recovery, flush area with water.

Small Spills:

Wipe up with absorbent material (e.g. cloth, fleece). Clean surface thoroughly to remove residual contamination.

Reference to other sections See also section 8 and 13

7. HANDLING AND STORAGE

Precautions for safe handling:

Avoid contact with skin and eyes, inhalation of vapors and mists.

Don't use empty container before they have been cleaned.

Before making transfer operations, assure that there aren't any incompatible material residuals in the containers.

Contaminated clothing should be changed before entering eating areas.

Do not eat or drink while working.

See also section 8 for recommended protective equipment.

Conditions for safe storage, including any incompatibilities

Keep away from food, drink and feed

Incompatible materials:

None in particular

Instructions as regards storage premises:

Adequately ventilated premises

Specific end use(s)

None in particular

8. EXPOSURE CONTROL/PERSONAL PROTECTION

Control parameters

DNEL Exposure Limit Values

Not available

PNEC Exposure Limit Values

Not available



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Personal Protective Equipment

Eyes Protection Requirements:

Not needed for normal use. Anyway, operate according good working practices.

Skin Protection Requirements:

No special precaution must be adopted for normal use.

Hand Protection Requirements:

Not needed for normal use. Precautionary wear gloves.

Respiratory Protection Requirements:

Not needed for normal use.

Thermal Hazards:

None

Environmental exposure controls:

None

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance and color:	Liquid
Odor:	Slight, characteristic
Odor threshold:	Not available
pH:	6.25 (1% in water) (CIPAC MT 75.3)
Melting point / freezing point:	< 0° C
Initial boiling point and boiling range:	> 100 °C
Vapor density:	Not available
Evaporation rate:	Not available
Vapor pressure:	0.14 mPa at 20 °C (ref. to tetraconazole) (OECD 104)
Relative density:	1.058 at 20 °C kg/l
Solubility in water:	It gives emulsion
Partition coefficient (n-octanol/water):	log P = 3.53 (ref. to tetraconazole) (OECD 107)
Viscosity:	Not available (on the basis of the components)

Other information

Miscibility:	Not available
Fat Solubility:	Not available
Conductivity:	Not available



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10. STABILITY AND REACTIVITY

Reactivity:

Stable under normal conditions

Chemical stability:

Stable under normal conditions

Possibility of hazardous reactions:

None

Conditions to avoid:

Stable under normal conditions

Incompatible materials:

None in particular

Thermal stability:

No decomposition if used according to specifications.

Hazardous decomposition products:

None

11. TOXICOLOGICAL INFORMATION

Toxicity class WHO (a.i.): II

Information on Acute toxicity for end product, Tetraconazole:

LD50 (Oral, rat): > 2,000 mg/kg

LD50 (Dermal, rabbit): > 2,000 mg/kg

LC50 (Inhalation, rat, 4h): 2.83 mg/L air (male, nose only)
> 3.66 mg/L air (max attainable conc.)

Skin irritation (rabbit): Not irritant.

Eye irritation (rabbit): Not irritant.

Skin sensitization: Not classified as sensitizer

Carcinogenicity data

No carcinogenic effect

Mutagenicity data

No mutagenic effect



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Reproductive/Teratology data

No teratogenic effect

Reproductive data

No evidence of reproduction toxicity

If not differently specified, the information required in Regulation 453/2010/EC listed below must be considered as not available:

- a) acute toxicity;
- b) skin corrosion/irritation;
- c) serious eye damage/irritation;
- d) respiratory or skin sensitisation;
- e) germ cell mutagenicity;
- f) carcinogenicity;
- g) reproductive toxicity;
- h) STOT-single exposure;
- i) STOT-repeated exposure;
- j) aspiration hazard.

12. ECOLOGICAL INFORMATION

Toxicity

Data referred to the mixture, Tetraconazole:

Fish – Acute toxicity:

Zebra fish (*Brachydanio rerio*),

LC50 (96h): > 100 mg/L (OECD 203)

Invertebrates – Acute toxicity:

Daphnia magna,

EC50 (48 h): 42.1 mg/L (OECD 202)

Algae – Acute toxicity:

Scenedesmus subspicatus,

EbC50 (72h): 10.4 mg/L (OECD 201)

ErC50 (72h): 39.7 mg/L (OECD 201)

Persistence and degradability

Data referred to Tetraconazole:

Water-sediment study:

DT50 (whole system): 310 – 372 g

DT50 (soil): 111.8 g (Geometric mean on studies 4 soils)



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Tetraconazole is stable to hydrolysis and is not expected its degradation by photolysis.

Tetraconazole is not readily degradable.

Bioaccumulative potential Data

Data referred to Tetraconazole:

BCF = 35.7 (whole fish)

Mobility in soil

Data referred to Tetraconazole:

Mobility: Koc from 531 to 1922 ml/g (4 soils)

II Tetraconazole ha low mobility in acid soils

Results of PBT and vPvB assessment

No PBT and/or vPvB in the mixture. No chemical safety report is required.

Other adverse effects

No other adverse effect to be mentioned.

13. DISPOSAL CONSIDERATIONS

Waste Disposal Methods:

Recover if possible. In so doing, comply with the local and national regulations currently in force.

14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

UN number:	3082
UN proper shipping name:	Not applicable
Transport hazard class:	9
Packing Group:	III
Environmental hazards:	Not applicable
Special Precautions for User:	Not applicable

15. REGULATORY INFORMATION

Classification of the substance or mixture

Directive criteria, 67/548/CE, 99/45/EC and following amendments thereof:

Symbol: Xn/Xi, N



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Indication of danger: Irritating substance

Label elements

Risk phrases:

R52/53 Harmful to aquatic organisms may cause long-term adverse effects in the aquatic environment.

Adverse physicochemical, human health and environmental effects:

No other hazards

Safety phrases:

S2 Keep out of reach of children.

S13 Keep away from food, drink and animal feeding stuffs.

S20/21 When using do not eat, drink or smoke.

S61 Avoid release to the environment. Refer to special instructions/Safety data sheets.

16. OTHER INFORMATION

R-phrases in section 3:

R20/22 Harmful by inhalation and if swallowed.

R36/38 Irritating to eyes and skin.

R51/53 Toxic to aquatic organisms may cause long-term adverse effects in the aquatic environment.

H-statements in section 3:

H411 Toxic to aquatic life with long lasting effects.

H302 Harmful if swallowed.

H332 Harmful if inhaled.

H315 Causes skin irritation.

H319 Causes serious eye irritation.